MATRIX GROUPS

GUNNAR MOGREN

ABSTRACT. Matrix groups are continuous symmetry groups. Since matrices can be identified with points in $\mathbb{R}^n$ we can talk about their topology. We will introduce tools such as the exponential map and Lie algebras. As an application we will show that $SO(3)$ is diffeomorphic to $\mathbb{RP}^3$.

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CONTENTS

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1. Introduction

Symmetry appears all over mathematics: from symmetries of an equation to geometric symmetries. Some of these symmetries can be continuous, such as rotations. Matrix groups are an algebraic way to study these continuous symmetries. And it turns out that matrix groups are manifolds! In this paper, we will consider matrix groups as groups of linear transformations, without first fixing a basis. This is not to say that using the matrices and fixing a basis is not important. Bases allow us to talk about the topology of matrix groups, and work algebraically with them. So after introducing them this way, we will move to working with matrix representations.
Matrix groups have many applications in physics, engineering, and mathematics.
We will begin by giving some group theory and matrix background. Then we will talk about famous matrix groups, and prove some algebraic properties we will need. Then we will talk about matrix group topology, once again proving some results
Our end goal is to prove the result that $SO(3)$ is diffeomorphic to $\mathbb{RP}^3$.

2. Background

2.1. **Groups.** Our central objects of the paper are groups, so we will begin by giving some group theory background.
A binary operation is a way to "combine" elements of a set into another element of the set. This will give us out group structure.

Definition 2.1. (Binary Operation on G)
Let $G \neq \emptyset$, then a binary operation is a function

$$\psi : G \times G \to G.$$

Remark 2.2. A group can be thought of as a set of symmetries of an object with an operation to compose these symmetries.

Definition 2.3. (Group)
A set G along with a binary operation $*$, so that:

(1) G is closed under the operation. That is $g * h \in G \ \forall g, h \in G$.
(2) G has an identity element e. That is $e * g = g * e = g \ \forall g \in G$.
(3) Every element of G has an inverse. That is $\forall g \in G \ \exists g^{-1} \in G$, $g * g^{-1} = g^{-1} * g = e$.
(4) The operation $*$ is associative. That is $(g * h) * j = g * (h * i)$ $\forall g, h, j \in G$.

An abelian group is a group where the operation is commutative.

Definition 2.4. (Abelian Group)
A group G such that

$$a * b = b * a, \ \forall a, b \in G.$$

A group homomorphism is a function from one group to another that preserves group structure.

Definition 2.5. (Group Homomorphism)
Let G and G' be groups with operations $*$ and $\cdot$, respectively. A function

$$\theta : G \to G'$$

is a group homomorphism if for all $g, h \in G$

$$\theta(g * h) = \theta(g) \cdot \theta(h).$$

Remark 2.6. While a group homomorphism preserves some structure, a map called a group isomorphism preserves all of the structure of the group. This means the groups are essentially equivalent.

Definition 2.7. (Isomorphism)
A bijective group homomorphism. Two groups, G, G' are isomorphic if there exists an isomorphism

$$\psi : G \to G'$$

between them.

Analogous to vector spaces and subspaces, groups also have subgroups:

Definition 2.8. (Subgroup)
If G is a group, then $H \subseteq G$ is a subgroup if it is also a group with the operation inherited from G.

It turns out that it suffices to show only three things to prove that H is a subgroup.

Theorem 2.9. *H is a subgroup of G if and only if*

(1) The identity element $e \in H$.
(2) If $a, b \in H$, then $a * b \in H$.
(3) If $a \in H$, then $a^{-1} \in H$.

For a group homomorphism, the kernel is the set of all things that get mapped to the identity.

Definition 2.10. (Kernel)
Let $\theta : G \to H$ be a group homomorphism.

$$\ker(\theta) := \{g \in G \mid \theta(g) = e_H\}.$$

The elements in the codomain where a homomorphism lands is called the image.

Definition 2.11. (Image)
Let $\theta : G \to H$ be a group homomorphism.

$$\mathrm{Im}(\theta) := \{h \in H \mid \text{there exists } g \in G \text{ such that } \theta(g) = h\}.$$

Theorem 2.12. *For a group homomorphism $\theta : G \to H$, θ's kernel is a normal subgroup of G, and θ's image is a subgroup of H.*

2.2. **Matrices.** We will need sets to take entries of a matrix from, so we will begin defining a few "nicely behaved" sets.

Definition 2.13. (Skew Field)
A set $\mathbb{K}$ with operations $+, \cdot$ such that

(1) $a \cdot (b+c) = a \cdot b + a \cdot c$ and $(b+c) \cdot a = b \cdot a + c \cdot a$ for any $a, b, c \in \mathbb{K}$.
(2) $\mathbb{K}$ is an abelian group under $+$. (With its identity denoted "0".)
(3) $\mathbb{K} \setminus \{0\}$ is a group under $\cdot$.

A vector space is taken over a field, but we can remove some structure and take it over a skew field instead. This is called:

Definition 2.14. (Left Vector Space)
A vector space W over a skew field $\mathbb{K}$. Which means scalar multiplication is not commutative. We define it to be on the left, i.e. for $w \in W$ and $\alpha \in \mathbb{K}$

$$\alpha \cdot (w) = \alpha w.$$

Definition 2.15. (Field)
A skew field where $\mathbb{K} \setminus \{0\}$ is abelian under $\cdot$.

Some examples are $\mathbb{R}$ and $\mathbb{C}$.
As you might notice multiplication for a skew field is not necessarily commutative. An example is the quaternions.

Definition 2.16. (Quaternions)

$$\begin{aligned}\mathbb{H} = &\{a + bi + cj + dk : a, b, c, d \in \mathbb{R},\ i^2 = j^2 = k^2 = -1,\ \text{and} \\ &i \cdot j = k, j \cdot k = i, k \cdot i = j, j \cdot i = -k, k \cdot j = -i, i \cdot k = -j\}\end{aligned}$$

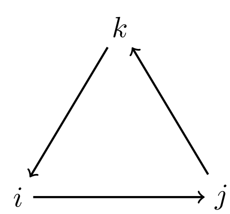

FIGURE 1. Quaternionic multiplication diagram

Figure 1 is a way to remember how quaternionic multiplication works. Multiplication is represented by the arrows, and going against them makes the product negative.

Notation 2.17. For this paper, $\mathbb{K}$ will denote $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$ unless stated otherwise.

Definition 2.18. ($M_n(\mathbb{K})$)
Let $\mathbb{K}$ be a skew field, the set of all $n \times n$ matrices with entries in a skew field $\mathbb{K}$ is denoted

$$M_n(\mathbb{K}).$$

Remark 2.19. $M_n(\mathbb{K})$ is not a group under matrix multiplication, as all its matrices do not have inverses. But it is a vector space under addition and scalar multiplication.

We now look at some operations on complex and quaternionic numbers and matrices.

Definition 2.20. (Conjugate of a Matrix)
The conjugate of a complex number, $z = a + bi$, is defined as

$$\overline{z} = a - bi.$$

The conjugate of a quaternionic number, $q = a + bi + cj + dk$, is defined as

$$\overline{q} = a - bi - cj - dk.$$

The conjugate of an $n \times n$ matrix, with either quaternionic or complex entries is defined as

$$\overline{A} = \begin{pmatrix} \overline{a_{11}} & \cdots & \overline{a_{1n}} \\ \vdots & \ddots & \vdots \\ \overline{a_{n1}} & \cdots & \overline{a_{nn}} \end{pmatrix}.$$

Definition 2.21. (Adjoint of a Matrix)
The adjoint of a matrix is defined as

$$\overline{A^T},$$

and denoted

$$A^*.$$

Definition 2.22. (Norm of a Matrix)
The norm of a matrix with entries in $\mathbb{K}$,

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix},$$

is defined by

$$||A|| = \sqrt{\sum_{l=1}^{n} \left(\sum_{k=1}^{n} |a_{lk}|^2 \right)}.$$

Now that we have our background, we will move on to the main objects of the paper.

3. Matrix Groups

3.1. **Definitions.** In this section we will study the algebraic structure of matrix groups (and a little topology). When first defining a specific matrix group, we will do so without fixing a basis, so we will be considering the elements as linear transformations and not necessarily matrices.

Why does it make sense to study the topology of matrices? Well, matrices can be identified with points in a higher dimensional Euclidean space.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longleftrightarrow (a, b, c, d)$$

So we can identify

$$M_n(\mathbb{K}) \longleftrightarrow \mathbb{K}^{n^2}.$$

Where $\mathbb{K}$ is a skew field. So we will identify the topology of $M_n(\mathbb{K})$ with the standard topology on $\mathbb{K}^{n^2}$.
For example, if $\mathbb{K} = \mathbb{R}$, then the open sets are unions of open balls in $\mathbb{R}^{n^2}$.
So now we encounter our first group.

Definition 3.1. ($GL(V)$)
Let V be a real vector space. The general linear group denoted $GL(V)$ is

$$GL(V) = \{A \mid A : V \to V \text{ is a linear isomorphism.}\}$$

Its group operation is composition.

We will want to consider the determinant function for the following definitions. So recall that the determinant of a linear transformation is not dependent on a choice of basis. The determinant is a function

$$\det : M_n(\mathbb{K}) \to \mathbb{K}.$$

Definition 3.2. (Determinant of a Linear Transformation)
Let $T : V \to V$ be a linear transformation of a vector space V over a field $\mathbb{K}$. Fix a basis of V, and let $[T]$ be the matrix that represents T in our basis. Then define

$$\det(T) = \det([T]).$$

Remark 3.3. Definition 3.2 is well defined because the determinant of a linear transformation is invariant under change of basis. If we have a linear transform T and a matrix representation after fixing a basis $[T]$, then if we wish to change basis we conjugate by a change of basis matrix C.

$$C[T]C^{-1}$$

Which we can take the determinant of this new matrix

$$\det(C[T]C^{-1}) = \det(C)\det([T])\det(C^{-1}) = \det(C)\det([T])\frac{1}{\det(C)} = \det([T]).$$

But after we fix a basis for V, we have a equivalent definition for the general linear group.

Definition 3.4. ($GL_n(\mathbb{K})$)
Let $\mathbb{K}$ be a skew field.

$$GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid \det(A) \neq 0\}$$

The group operation is matrix multiplication.

Proof. We will prove that $GL_n(\mathbb{K})$ is a group, (with $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$). Let $A, B \in GL_n(\mathbb{K})$ be given. Then

$$\det(AB) = \det(A)\det(B).$$

We know that $\det(A) \neq 0$ and $\det(B) \neq 0$ so

$$\det(AB) \neq 0.$$

So, $GL_n(\mathbb{K})$ is closed under matrix multiplication. Its identity element is the identity matrix, all its matrices are invertible, and matrix multiplication is associative.

□

Remark 3.5. The determinant is defined differently when $\mathbb{K} = \mathbb{H}$, but the above argument still works for the quaternionic determinant.

The group $GL_n(\mathbb{K})$ is open in $M_n(\mathbb{K})$ (which we will prove in the next chapter) and will inherit the subspace topology from $\mathbb{K}^{n^2}$. We are now ready for the definition of a matrix group.

Definition 3.6. (Matrix group)
A closed subgroup of $GL_n(\mathbb{K})$..

Matrix groups are Lie groups, which are defined as:

Definition 3.7. (Lie Group)
A group G which is a manifold, the binary operation on G

$$\mu : G \times G \to G$$

is smooth, and the inverse map

$$g \mapsto g^{-1}$$

is smooth.

This also changes out definition of homomorphism slightly, so we have the definitions:

Definition 3.8. (Lie Group Homomorphism)
A group homomorphism that is also smooth.

Definition 3.9. (Lie Group Isomorphism)
A group isomorphism that is also smooth.

So now we will introduce some common matrix groups besides $GL(V)$. The first that we will consider the the special linear group. Its members are linear isomorphisms that, from a geometric perspective, preserve area (or any higher dimensional analog) in a vector space.

Definition 3.10. ($SL(V)$)
Let V be a real vector space.

$$SL(V) = \{A \in GL(V) \mid \det(A) = 1\}$$

Notice that this definition is still valid without a basis, because the determinant of a linear transformation is not dependent on the choice of basis. After fixing a basis, the definition is still that its determinant is 1.
Now we will consider is the orthogonal group. Members of the orthogonal group are linear isomorphisms that, from a geometric perspective, preserve angles and distance. We can think about these transformations as rotations and reflections.

Definition 3.11. ($O(V)$)
Let V be a vector space.

$$O(V) = \{A \mid A : V \to V \text{ is a linear isomorphism and } \forall v, u \in V, \langle Av, Au\rangle_{\mathbb{R}} = \langle v, u\rangle_{\mathbb{R}}\}$$

Where $\langle -, -\rangle_{\mathbb{R}}$ is the standard inner product for a real vector space.

Some equivalent definitions, after fixing a basis for V, are:

Definition 3.12. ($O_n(\mathbb{K})$)

(1) $O_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) \mid AA^T = I\}$
(2) $O_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) \mid \text{The columns of } A \text{ are orthonormal.}\}$

The next is the special orthogonal group. The geometric way we can think about this group, is that it preserves angles and distances, and also orientation.

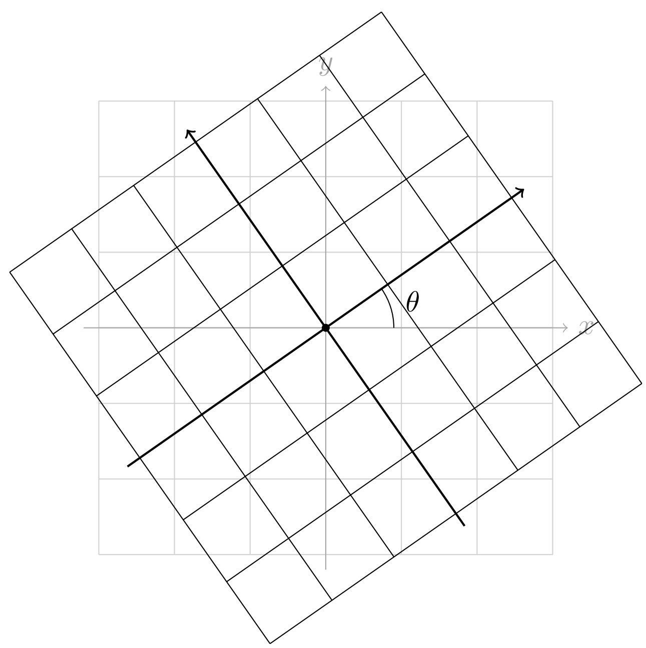

FIGURE 2. An example of how a member of $SO(\mathbb{R}^2)$ might transform the plane.

Definition 3.13. ($SO(V)$)
Let V be a vector space.

$$SO(V) = \{A \in O(V) \mid \det(A) = 1\}$$

Once we fix a basis we have the definition:

Definition 3.14. ($SO_n(\mathbb{K})$)

$$SO_n(\mathbb{K}) = \{A \in O_n(\mathbb{K}) \mid \det(A) = 1\}$$

The next matrix group we will consider is the unitary group. To talk about this group, we will also need an additional definition.

Definition 3.15. (Hermitian Inner Product)
Let u, v be in an n dimensional complex vector space V. Then the hermitian inner product is

$$\langle u, v\rangle_{\mathbb{C}} = \sum_{l=1}^{n} \overline{u_l} v_l.$$

Then you can think of the unitary group as a complex version of the orthogonal group. So it will have additional structure. The unitary group can be defined over any skew field, but in this paper, we will define it over a complex vector space.

Definition 3.16. ($U(V)$)
Let V be a vector space.

$$U(V) = \{A \mid A : V \to V \text{ is a linear isomorphism and } \forall u, v \in V, \langle Au, Av\rangle_{\mathbb{C}} = \langle u, v\rangle_{\mathbb{C}}\}$$

After fixing a basis for V we have the equivalent definitions:

Definition 3.17. ($U_n(\mathbb{C})$)

(1) $U_n(\mathbb{C}) = \{A \in M_n(\mathbb{C}) \mid AA^* = I\}$
(2) $U_n(\mathbb{C}) = \{A \in M_n(\mathbb{C}) \mid |\det(A)| = 1\}$

The special unitary group is again analogous to the special orthogonal group, but has extra complex structure.

Definition 3.18. ($SU(V)$)
Let V be a vector space.

$$SU(V) = \{A \in U(V) \mid \det(A) = 1\}$$

After fixing a basis, we have the equivalent definitions.

Definition 3.19. ($SU_n(\mathbb{C})$)
$SU_n(\mathbb{C}) = \{A \in U_n(\mathbb{C}) \mid \det(A) = 1\}$

Geometrically, we can once again think about these last two groups as preserving angles and distance. But in complex geometry instead of Euclidean. For the next group we will need the following definition.

Definition 3.20. (Quaternionic Inner Product)
Let u, v be in an n dimensional quaternionic left vector space V. Then the quaternionic inner product is

$$\langle u, v\rangle_{\mathbb{H}} = \sum_{l=1}^{n} \overline{u_l} v_l.$$

Finally, the symplectic group can be thought of as the quaternionic analog of the orthogonal group. Once again preserving angles and distances, but in quaternionic geometry. There are actually multiple definitions of the symplectic group in mathematics, and they are not all equivalent. But for this paper, we will use the following definition.

Definition 3.21. ($Sp(W)$)
Let W be a quaternionic left vector space. Then

$$Sp(W) = \{B \mid B : W \to W \text{ is a linear isomorphism and } \forall u, v \in W, \langle Bu, Bv\rangle_{\mathbb{H}} = \langle u, v\rangle_{\mathbb{H}}\}$$

3.2. **All Matrix Groups are Real Matrix Groups.** We are going to delay the equivalent definitions of the symplectic group because we must look at some representation theory. The determinant behaves differently for quaternionic matrices, so we need a different definition. It turns out that every matrix group is isomorphic to a subgroup of $GL(\mathbb{R}^n)$, for some n. To show this we will define two maps.

Definition 3.22. First recall that any complex number can be written as $z = a + bi$. We consider the function $\rho_n : M_n(\mathbb{C}) \to M_{2n}(\mathbb{R})$ where

$$\rho_1(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

and then

$$\rho_n(\begin{pmatrix} z_{11} & \cdots & z_{1n} \\ \vdots & \ddots & \vdots \\ z_{n1} & \cdots & z_{nn} \end{pmatrix}) = \begin{pmatrix} \rho_1(z_{11}) & \cdots & \rho_1(z_{1n}) \\ \vdots & \ddots & \vdots \\ \rho_1(z_{n1}) & \cdots & \rho_1(z_{nn}) \end{pmatrix}.$$

Restricting ρ_n's domain to a subgroup of $M_n(\mathbb{C})$ and its codomain to its image in $M_{2n}(\mathbb{R})$ induces an isomorphism. But we will not prove that here, for reference see [Tap05, Ch. 2].

Definition 3.23. Recall that any quaternion can be written as
$q = a + bi + cj + dk = (a + bi) + (c + di)j$. We define the function
$\Psi_n : M_n(\mathbb{H}) \to M_{2n}(\mathbb{C})$ where

$$\Psi_1((a + bi) + (c + di)j) = \begin{pmatrix} a + bi & c + di \\ -(\overline{c + di}) & \overline{a + bi} \end{pmatrix}$$

and then

$$\Psi_n(\begin{pmatrix} q_{11} & \cdots & q_{1n} \\ \vdots & \ddots & \vdots \\ q_{n1} & \cdots & q_{nn} \end{pmatrix}) = \begin{pmatrix} \Psi_1(q_{11}) & \cdots & \Psi_1(q_{1n}) \\ \vdots & \ddots & \vdots \\ \Psi_1(q_{n1}) & \cdots & \Psi_1(q_{nn}) \end{pmatrix}.$$

If Ψ_n is restricted to a subgroup of $M_n(\mathbb{H})$, then once again it is isomorphic to its image, see [Tap05, Ch. 2]. We are now ready to define the quaternionic determinant.

Definition 3.24. (Quaternionic Determinant)
If $A \in M_n(\mathbb{H})$, we define

$$\det_{\mathbb{H}}(A) = \det(\Psi(A)).$$

3.3. **Closing Remarks.** Now we are ready to give the equivalent definitions for the symplectic group.

Definition 3.25. ($Sp_n(\mathbb{H})$)

(1) $Sp_n(\mathbb{H}) = \{A \in M_n(\mathbb{H}) \mid AA^* = I\}$
(2) $Sp_n(\mathbb{H}) = \{A \in M_n(\mathbb{H}) \mid |\det_{\mathbb{H}}(A)| = 1\}$

Also, for $GL_n(\mathbb{H})$ the definition is analogous to definition 4.3, and the proof it is a group would be the same but with the quaternionic determinant. Finally, we will discuss some alternative notation.

Notation 3.26.

(1) $O(n)$ and $SO(n)$ denote $O_n(\mathbb{R})$ and $SO_n(\mathbb{R})$ respectively.
(2) $U(n)$ and $SU(n)$ denote $U_n(\mathbb{C})$ and $SU_n(\mathbb{C})$ respectively.
(3) $Sp(n)$ denotes $Sp_n(\mathbb{H})$.

Now that we have been introduced to the matrix groups we will be studying, we will look at their topological properties.

4. Topology of Matrix Groups

In this section we will study the topology of matrix groups using their algebraic properties from the last section.

Theorem 4.1. *$GL_n(\mathbb{K})$ is open in $M_n(\mathbb{K})$.*

Proof. Consider the first case when $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. The determinant function is a map

$$\det : M_n(\mathbb{K}) \to \mathbb{K}.$$

Then consider that

$$\det^{-1}(\mathbb{K} \setminus \{0\}) = GL_n(\mathbb{K}).$$

Since the determinant is continuous, the preimage of an open set is open. Then the other case: $\mathbb{K} = \mathbb{H}$. The determinant then maps

$$\det_{\mathbb{H}} : M_n(\mathbb{H}) \to \mathbb{C}.$$

So we have
$$\det_{\mathbb{H}}{}^{-1}(\mathbb{H} \setminus \{0\}) = GL_n(\mathbb{H}).$$
Hence, $GL_n(\mathbb{K})$ is open.

□

Interestingly, $GL_n(\mathbb{R})$ is not path connected. This is because its image is
$$\mathbb{R} \setminus \{0\}.$$
So we have
$$GL_n(\mathbb{R}) = \det^{-1}((-\infty), 0) \cup \det^{-1}((0, \infty)),$$
and the pre-image of the image of a continuous map is disconnected if the image is disconnected. But interestingly $GL_n(\mathbb{C})$ is path connected, because $\mathbb{C} \setminus \{0\}$ is not disconnected.

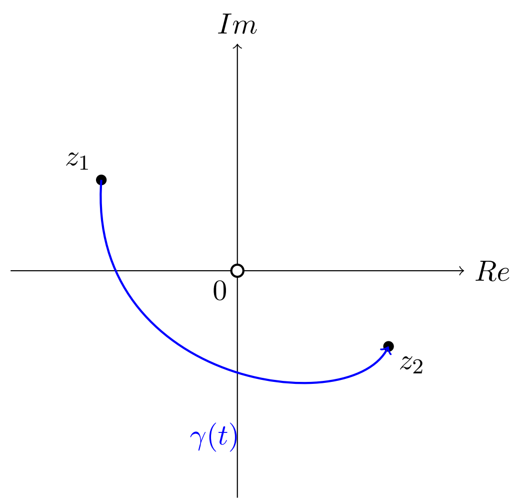

Figure 3. A path in $\mathbb{C} \setminus \{0\}$ avoiding the origin.

We now look at some specific homeomorphisms.

Theorem 4.2.
$SO(2)$ is homeomorphic to S^1.

Proof. For S^1, we will use the definition in the complex plane:
$$S^1 = \{z \in \mathbb{C} : |z| = 1\}$$
It is known that the elements of $SO(2)$ are in the form
$$\begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$
And any complex number, z, can be written as
$$re^{i\theta}.$$

So the definition of S^1 can be

$$S^1 = \{e^{i\theta} : \theta \in [0, 2\pi)\},$$

Define a function $\psi : S^1 \to SO(2)$ as

$$\psi(e^{i\theta}) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

This function is bijective and continuous. And because its domain is compact, and its codomain is Hausdorff (this is because $\mathbb{R}^4$ is Hausdorff), it is a homeomorphism. □

We now consider:

Theorem 4.3.

$Sp(1)$ is homeomorphic to S^3.

Proof. Every $q \in Sp(1)$ has the property $q \cdot \overline{q} = 1$ which implies that $|q| = 1$. So an equivalent definition for $Sp(1)$ is

$$Sp(1) = \{a + bi + cj + dk \in \mathbb{H} : \sqrt{a^2 + b^2 + c^2 + d^2} = 1\}.$$

Consider the function $\psi : Sp(1) \to S^3$ with

$$\psi(a + bi + cj + dk) = (a, b, c, d).$$

This function is bijective and continuous. And because its domain is compact, and its codomain is Hausdorff, it is a homeomorphism.

□

Corollary 4.4. *$Sp(1)$ is compact and path connected.*

Theorem 4.5. *$SO(n)$ is path connected.*

First we will need the lemma:

Lemma 4.6. *Given $v \in \mathbb{R}^n$ with $|v| = 1$ there is a path, $A_t : [0, 1] \to SO(n)$, in $SO(n)$ such that*

$$A_0 = I \text{ and } A_1 v = e_1$$

where e_1 is the unit vector

$$(1, 0, 0, \cdots)$$

on S^{n+1}.

Now we are ready for the proof of theorem 4.5.

Proof. It will be sufficient to show that there is a path from the identity to any element of the group.
Given $R \in SO(n)$ with $v := Re_1$. This means $|v| = 1$. By lemma 4.6 we have that $A_t \in SO(n)$ such that $A_0 = I$ and $A_1 v_1 = e_1$. Now consider the path

$$\gamma : [0, 1] \to SO(3) \text{ where } \gamma(t) = A_t R$$

with $\gamma(0) = A_0 R = R$ and also

$$\gamma(1) = A_1 R \text{ and } A_1 R e_1 = A_1 v_1 = e_1.$$

This implies that $\gamma(1)$ is a block matrix in the form:

$$\gamma(1) = \begin{pmatrix} 1 & 0 \\ 0 & R_1 \end{pmatrix}$$

with $R_1 \in SO(n)$. Apply the same argument to R_1 to get

$$\gamma_2 : \gamma(1) \mapsto \begin{pmatrix} 1 & & \\ & 1 & \\ & & R_1 \end{pmatrix}.$$

And then by induction:

$$\gamma_n : \gamma(n-1) \mapsto \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}.$$

Then we have the path

$$\alpha = \gamma_n \circ \cdots \circ \gamma_2 \circ \gamma,$$

and $\alpha(0) = R$ and $\alpha(1) = I$.

□

We will need a powerful tool to help us explore more topological properties of matrix groups. Which we will introduce in the next section.

5. The Matrix Exponential

The matrix exponential will allow us to eventually work with the Lie algebras of matrix groups. But we will need some properties first.
Recall that the power series for the exponential function is

$$\exp(x) = e^x = \sum_{i=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$

Which has radius of convergence $(-\infty, \infty)$. So we define the exponential of a matrix as

Definition 5.1. For a matrix $A \in M_n(\mathbb{K})$

$$\exp(A) = e^A = \sum_{i=0}^{\infty} \frac{A^n}{n!} = 1 + \frac{A}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots.$$

This converges for any matrix, which we will not prove here. For reference see [Tap05, Ch. 6]. Matrix exponentiation does not behave exactly like normal exponentiation, so we will discuss its properties.

Proposition 5.2. *For $A, B \in M_n(\mathbb{K})$, if $AB = BA$, then*

$$e^A \cdot e^B = e^{A+B}.$$

Notice that the above proposition only works if the matrices commute.

Proof.

$$\begin{aligned}
e^A e^B &= \left(\sum_{l=0}^{\infty} \frac{A^l}{l!}\right)\left(\sum_{k=0}^{\infty} \frac{B^k}{k!}\right) \\
&= \sum_{l=0}^{\infty}\sum_{k=0}^{l} \frac{A^k B^{l-k}}{k!(l-k)!} \\
&= \sum_{l=0}^{\infty} \frac{1}{l!}\sum_{k=0}^{l} \binom{l}{k} A^k B^{l-k} \\
&= \sum_{l=0}^{\infty} \frac{(A+B)^l}{l!} \\
&= e^{A+B}
\end{aligned}$$

□

To see reference for the summation properties used, see [Tap05, Ch. 6]. Another property is:

Theorem 5.3. *If $A \in M_n(\mathbb{K})$ with $\mathbb{K} \in \{\mathbb{C}, \mathbb{R}\}$, then*

$$\det(e^A) = e^{tr(A)}.$$

See [Tap05, p. 88].
Our last property we will look at is:

Proposition 5.4. *Let $A, B \in M_n(\mathbb{K})$ with A invertible. Then*

$$e^{ABA^{-1}} = Ae^B A^{-1}.$$

Proof.

$$\begin{aligned}
Ae^B A^{-1} &= A\left(I + B + \frac{1}{2!}B^2 + \frac{1}{3!}B^3 + \cdots\right)A^{-1} \\
&= I + ABA^{-1} + \frac{1}{2!}AB^2A^{-1} + \frac{1}{3!}AB^3A^{-1} + \cdots \\
&= I + ABA^{-1} + \frac{1}{2!}(ABA^{-1})^2 + \frac{1}{3!}(ABA^{-1})^2 + \cdots \\
&= e^{ABA^{-1}}
\end{aligned}$$

□

Now that we are equipped with this powerful tool, we will go on to study the topology of matrix groups more.

6. Matrix Groups are Manifolds

The goal of this section is to prove that matrix groups are manifolds. But first we want to give a definition for tangent spaces.

Definition 6.1. (Tangent Space)
The tangent space at a point p on an open subset $M \subseteq \mathbb{R}^n$ is defined as

$$T_pM = \{\gamma'(0) \mid \gamma : (-\varepsilon, \varepsilon) \to M, \gamma(0) = p\}.$$

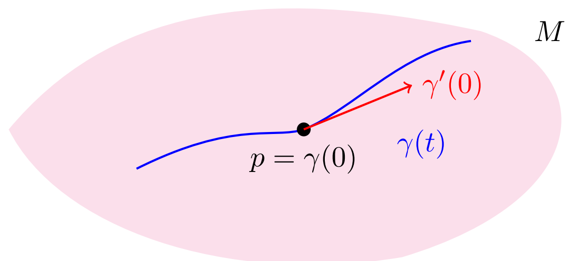

FIGURE 4. A path γ on a manifold M, with its derivative at p.

We will give a temporary definition for Lie algebras, to assist us this chapter.

Definition 6.2. (Lie Algebra)
The tangent space for a matrix group G at the identity is the Lie algebra of the group, or

$$\mathfrak{g} = T_IG.$$

Also the following notation will be useful.

Notation 6.3. Let $r > 0$

$$B_r = \{A \in M_n(\mathbb{K}) \ : \ ||A|| < r\}$$

We will use a theorem proved in [Tap05, Ch. 4].

Theorem 6.4. *If G is a matrix group with corresponding Lie algebra $\mathfrak{g}$*

(1) For $X \in \mathfrak{g}$, $e^X \in G$.
(2) For sufficiently small $r > 0$,

$$V = \exp(B_r \cap \mathfrak{g})$$

is a neighborhood of I in G, and

$$\exp : B_r \cap \mathfrak{g} \to V$$

is a diffeomorphism.

Result (1) of the above theorem allows us to construct a path on any matrix group through the identity! This will greatly assist us in computing the Lie algebras of matrix groups. Then we are ready to for the main theorem of the chapter

Theorem 6.5. *Every matrix group is a smooth manifold.*

Proof. By theorem 7.4 part (2), G has a parametrization around I. We define a function for $g \in G$

$$L_g : G \to G, \text{ with } L_g(h) = gh.$$

This function is smooth and has a smooth inverse $L_{g^{-1}}$. So L_g is a diffeomorphism. Then we consider the function from theorem 7.4 part (2) and compose it with L_g:

$$L_g\big|_V \circ \exp\big|_{B_r} : B_r \cap \mathfrak{g} \to L_g(V)$$

A smooth function composed with another smooth function is smooth. Also, since L_g is restricted to V, is continuous, and $L_g(I) = g$, this "moves" V to a neighborhood of g. This allows us to parametrize the whole group. So

$$L_g\big|_V \circ \exp\big|_{B_r}$$

is a diffeomorphism. Hence all matrix groups are smooth manifolds.

□

So since smooth manifolds are defined to have the same dimension as their tangent space. We will define

Definition 6.6. (Dimension of a Matrix Group)
The dimension of a matrix group is the dimension of its Lie algebra as a vector space.

7. Lie Algebras

The goal of this section is to explain what a Lie algebra is, where its structure comes from, and prove some results that we will use to prove our final result of the paper.
Now we will give the real definition of a Lie algebra, instead of the temporary one given in the previous section.

Definition 7.1. (Lie Algebra)
A vector space V with a bilinear operation

$$[-,-] : V \times V \to V$$

that satisfies

(1) Anti-symmetry: $[X, Y] = -[Y, X]$
(2) Jacobi Identity: $[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$

for all $X, Y, Z \in V$. This bilinear operation is called the Lie bracket, and for a Lie group G, its corresponding Lie algebra equipped with the Lie bracket, will be the vector space $V = T_I G$.

So since we have added more structure to a vector space, we will need a new isomorphism definition.

Definition 7.2. (Lie Algebra Isomorphism)
If $\mathfrak{g}$ and $\mathfrak{h}$ are Lie algebras, and

$$T : \mathfrak{g} \to \mathfrak{h}$$

is a linear isomorphism. If for $X, Y \in \mathfrak{g}$

$$T([X, Y]_{\mathfrak{g}}) = [T(X), T(Y)]_{\mathfrak{h}},$$

then T is a Lie algebra isomorphism.

We will need the following definitions to talk about the Lie bracket for matrix groups.

Definition 7.3. (Conjugation by g)
Let G be a matrix group. For $g \in G$, $C_g : G \to G$ and for $h \in G$

$$C_g(h) = ghg^{-1}.$$

Then we want to compute C_g's derivative at the identity. We do this by considering a path γ through the identity, and conjugating it. Let

$$\gamma : (-\varepsilon, \varepsilon) \to G, \text{ with } \gamma(0) = I \text{ and } \gamma'(0) = X \text{ for some } X \in \mathfrak{g}.$$

Then we have

$$d(C_g)\big|_I = \frac{d}{dt}\big|_{t=0} g\gamma(t)g^{-1} = gXg^{-1}.$$

We define the derivative of C_g at the identity as the adjoint action.

Definition 7.4. (Adjoint Action)
First we have the map

$$\mathrm{Ad} : G \to GL(\mathfrak{g}).$$

(Recall that $GL(\mathfrak{g})$ is the group of linear isomorphisms of $\mathfrak{g}$.) Once you fix a $g \in G$, we have the map

$$\mathrm{Ad}_g : \mathfrak{g} \to \mathfrak{g}.$$

So for $X \in \mathfrak{g}$, we have

$$\mathrm{Ad}_g(X) = gXg^{-1}.$$

We will need the following property for the next chapter.

Theorem 7.5. *$Ad : G \to GL(\mathfrak{g})$ is smooth.*

The above theorem is because it is differentiable.
Now we consider the derivative of the adjoint action at the identity. We will define this as

Definition 7.6. (Infinitesimal Adjoint Action)
A map

$$\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}).$$

Recall that $\mathfrak{gl}(\mathfrak{g})$ is the vector space of endomorphisms of $\mathfrak{g}$. So once you fix an $X \in \mathfrak{g}$ and are given $Y \in \mathfrak{g}$

$$\mathrm{ad}_A(X) = \mathrm{d}(\mathrm{Ad}_{e^{tX}})\big|_{t=0}.$$

Definition 7.7. (Lie Bracket for a General Lie Group)
The Lie bracket, for a general Lie group's Lie algebra $\mathfrak{g}$, is defined as

$$[X, Y] := \mathrm{ad}_X(Y)$$

It turns out that for a matrix group, the Lie bracket is the commutator.

Theorem 7.8. *For a matrix group G, the Lie bracket for its Lie algebra is*

$$[X, Y] = XY - YX.$$

Proof. Let G be a matrix group with Lie algebra $\mathfrak{g}$. Around the identity every $g \in G$ can be written as

$$g = e^X \text{ for some } X \in \mathfrak{g}.$$

We consider the adjoint action

$$\mathrm{Ad}_{e^{tX}}(Y) = e^{tX} Y e^{-tX}.$$

And we take its derivative at the identity:

$$\begin{aligned} \frac{d}{dt}\Big|_{t=0} \mathrm{Ad}_{e^{tX}}(Y) &= \frac{d}{dt}\Big|_{t=0} e^{tX} Y e^{-tX} \\ &= (Xe^{tX} Y e^{-tX} - e^{tX} Y X e^{-tX})\Big|_{t=0} \\ &= XY - YX \end{aligned}$$

□

We will need a specific property of the adjoint action for the next chapter.

Theorem 7.9. *If the fixed basis $\mathcal{B}$ of $\mathfrak{g}$ is orthonormal with respect to $\langle -, - \rangle_{\mathbb{R}}$, then $Ad_g \in O(n)$ for all $g \in G$. Thus,*

$$Ad : G \to O(n)$$

is a homomorphism into the orthogonal group.

For reference see [Tap05, p. 123].
Now we will look at some examples of Lie algebras of the matrix groups defined in section 4.

Theorem 7.10. *The Lie algebra of $GL_n(\mathbb{K})$, denoted $\mathfrak{gl}_n(\mathbb{K})$, is $M_n(\mathbb{K})$.*

Proof. Consider the path through the identity for $GL_n(\mathbb{K})$.

$$\gamma : (-\varepsilon, \varepsilon) \to GL_n(\mathbb{K}), \text{ with } \gamma(t) = I + tA, \text{ where } A \in M_n(\mathbb{K}).$$

This gives

$$\gamma(0) = I \text{ and } \gamma'(0) = A.$$

Since, the determinant of I is 1, for sufficiently small t, the determinant of $\gamma(t)$ will always be non-zero. So our path always lands in $GL_n(\mathbb{K})$, and its derivative can be any $A \in M_n(\mathbb{K})$.

□

Now we will consider the Lie algebra for the special linear group.

Theorem 7.11.

$$\mathfrak{sl}_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid tr(A) = 0\}$$

For a proof of the above theorem, see [Tap05, p. 72]. Now we will look at the Lie algebra for the orthogonal group and the special orthogonal group.

Theorem 7.12.

$$\mathfrak{o}_n(\mathbb{K}) = \mathfrak{so}_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid A + A^* = 0\}$$

To see a proof of this theorem, see [Hal15, p. 40]. Geometrically, this makes sense because $O(n)$ has two connected components, and $SO(n)$ is exactly the one of those components that contains the identity.
Now we will look at a few more specific examples, which will help us with the next chapter.

Theorem 7.13. *The Lie algebra of $Sp(1)$, denoted $\mathfrak{sp}(1)$, is* $\mathrm{span}\{i, j, k\}$.

Geometrically, since $Sp(1) \cong S^3$, the Lie algebra can be thought of as the tangent space at $(1, 0, 0, 0)$ on the "surface" of the 3-sphere, i.e.

$$\mathrm{span}\{(0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)\}.$$

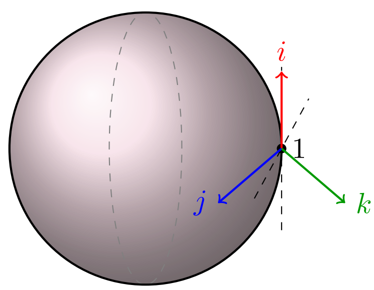

FIGURE 5. The Lie algebra $\mathfrak{sp}(1)$ as the tangent space at 1.

Proof. Consider the smooth path $\gamma : (-\varepsilon, \varepsilon) \to Sp(1)$ defined as

$$\gamma(t) = a(t) + b(t)i + c(t)j + d(t)k$$

with $q(0) = 1$. This implies that $a(0) = 1$ and $b(0) = c(0) = d(0) = 0$. By definition

$$q(t) \cdot q(t)^* = 1.$$

We differentiate

$$\frac{d}{dt}(q(t) \cdot q(t)^*) = 0$$
$$q'(t) \cdot q(t)^* + q(t) \cdot q'(t)^* = 0$$

Then using $q(0) = 1$ we have

$$q'(0) \cdot q(0)^* + q(0) \cdot q'(0)^* = q'(0) + q'(0)^* = 0.$$

Then consider that

$$q'(0) = a'(0) + b'(0)i + c'(0)j + d'(0)k$$
$$q'(0)^* = a'(0) - b'(0)i - c'(0)j - d'(0)k.$$

Their sum should be zero by our derivation, so

$$2a'(0) = 0 \implies a'(0) = 0.$$

Hence,

$$\mathfrak{sp}(1) = \operatorname{span}\{i, j, k\}.$$

□

Because, $\dim(\mathfrak{sp}(1)) = 3$, we have:

Corollary 7.14.

$$\mathfrak{sp}(1) \cong \mathbb{R}^3.$$

8. $SO(3)$ is diffeomorphic to $\mathbb{RP}^3$

8.1. **What is Real Projective Space?** The definition we will use for real projective space is:

Definition 8.1 (Real Projective Space)**.**

$$\mathbb{RP}^n = S^n/(x \sim -x)$$

This can be thought of as identifying antipodal points on an n-sphere.

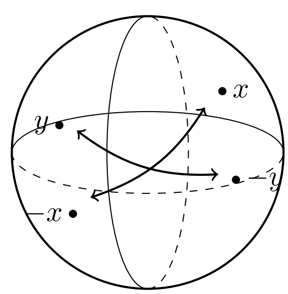

Figure 6. Identifying antipodal points on the 2-sphere.

But an equivalent definition is also:

Definition 8.2 (Real Projective Space)**.**

$$\mathbb{RP}^n = \mathbb{R}^{n+1}/(x \sim \lambda x)$$

with $\lambda \in \mathbb{R}$ and $x \in \mathbb{R}^{n+1}$.

The equivalence classes of the above quotient can be thought of as lines through the origin.

8.2. **The Main Result.** The goal of this section is to prove that $SO(3)$ is diffeomorphic to $\mathbb{RP}^3$. We will be considering the map

$$\mathrm{Ad} : Sp(1) \to GL(\mathfrak{sp}(1)).$$

Because of corollary 7.14 we can consider the map as

$$\mathrm{Ad} : Sp(1) \to GL(\mathbb{R}^3).$$

Then we can restrict the codomain of our map using theorem 7.9, to:

$$\mathrm{Ad} : Sp(1) \to O(3)$$

Since $O(3)$ has 2 path components and one of those components is $SO(3)$ (See [Hal15, Section 1.4]), we can further restrict it using corollary 4.4, theorem 7.5, and that
$\mathrm{Ad}(1) = I$ to:

$$\mathrm{Ad} : Sp(1) \to SO(3).$$

We will need the following property:

Theorem 8.3.

$$Ad : Sp(1) \to SO(3) \textit{ is a local diffeomorphism at } I.$$

See [Tap05, p. 127].

Corollary 8.4. *$SO(3)$ is open.*

We will also need more abstract algebra to assist us.

Theorem 8.5. *(First Isomorphism Theorem for Lie Groups)*
Let G and G' be groups, and

$$\psi : G \to G'$$

be a surjective Lie group homomorphism. Then

$$G/\ker(\psi) \cong G'.$$

Because the last theorem requires the homomorphism to be surjective, now our goal is to show that Ad surjective.

Theorem 8.6.

$$Ad : Sp(1) \to SO(3) \textit{ is surjective.}$$

Proof. Since $Sp(1)$ is compact and Ad is continuous, its image is compact. So it is closed by the Heine Borel theorem. But it is also open by theorem 8.3. Since $SO(3)$ is path connected its only closed and open sets are the empty set and itself, and the image of Ad is non-empty. Hence,

$$\mathrm{Ad} : Sp(1) \to SO(3)$$

is surjective. □

We also need the kernel of the adjoint action:

Theorem 8.7. *For the map* $Ad\colon Sp(1) \to GL(\mathfrak{sp}(1))$,

$$\ker(Ad) = \{\pm 1\}.$$

Proof. Let $s \in \mathfrak{sp}(1)$ and let $q \in Sp(1)$ such that

$$\mathrm{Ad}(q) = Ad_q.$$

We want to find out for which q, this is the identity function. So consider

$$Ad_{a+bi+cj+dk}(s) = qsq^{-1} = s$$

For this to be true we need q and q^{-1} to commute with s. And because any imaginary quaternion will not commute with s, q must only have real parts. Remember that $|q| = 1$. Hence,

$$\ker(\mathrm{Ad}) = \{\pm 1\}.$$

$\square$

We will also need the theorem:

Theorem 8.8.

$$Sp(1)/\{\pm 1\} \textit{ is } \mathbb{R}P^3.$$

Proof. Since $Sp(1)$ is S^3, then the quotient

$$Sp(1)/\{\pm 1\} = S^3/(x \sim -x).$$

This is exactly the definition of $\mathbb{R}\mathrm{P}^3$ so

$$Sp(1)/\{\pm 1\} = \mathbb{R}\mathrm{P}^3.$$

$\square$

We now have all the results necessary.

Theorem 8.9.

$$SO(3) \textit{ is diffeomorphic to } \mathbb{R}P^3.$$

Proof. Since the map $\mathrm{Ad} : Sp(1) \to SO(3)$ is a surjective Lie group homomorphism, and its kernel is $\{\pm 1\}$, we have that

$$Sp(1)/\{\pm 1\} \cong SO(3).$$

Then since $Sp(1)/\{\pm 1\}$ is $\mathbb{R}\mathrm{P}^3$ and that a Lie group isomorphism is a diffeomorphism,

$$SO(3) \text{ is diffeomorphic to } \mathbb{R}\mathrm{P}^3.$$

$\square$

8.3. **Closing Remarks.** Theorem 8.9 has a geometric interpretation. For an element of $SO(3)$, the axis of rotation can be thought of as a vector, $\vec{v}$, emanating from the origin. But the rotation does not depend on if $\vec{v}$ is scaled in any way. So every rotation of $\mathbb{R}^3$ can be identified with an ordered pair

$$(\vec{v}, \theta), \text{ where } v \in \mathbb{R}^3.$$

The ordered pair $(\vec{v}, \theta)$ can be identified with a point on S^3. And the same rotation can be represented as

$$(-\vec{v}, -\theta).$$

So

$$S^3/((\vec{v}, \theta) \sim (-\vec{v}, -\theta)) = \mathbb{R}\mathrm{P}^3.$$

References

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[Tap05] Kristopher Tapp, *Matrix groups for undergraduates*, Student Mathematical Library, vol. 29, American Mathematical Society, Providence, RI, 2005.